Response to Reviewers

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"An Empirical Study of Deep Reinforcement Learning for Dynamic Scheduling in Satellite QKD Networks"

We thank the Associate Editor and both reviewers for their careful and constructive reviews. We have revised the manuscript thoroughly; every point is addressed below. Reviewer comments are shown in shaded boxes, followed by our response. For reference, the principal changes are: a fully specified reward with a provably fair single-objective comparison, a switching-cost sensitivity analysis, a cost-aware hybrid baseline, multi-seed (20-seed) evaluation with mean and standard deviation, a correction to the switching-cost model, and an expanded motivation and Limitations section. All code and models are at `https://github.com/ailabteam/AILET`.

Associate Editor

> We see promise in this paper, but there is still work to be done to clarify the issues raised by the more negative of our reviewers.

Response. We have focused the revision on exactly these issues. The most negative reviewer (Reviewer 2) questioned whether the comparison was fair given an undisclosed reward. We now specify the reward in full and make explicit that all policies are scored by one identical physical objective (the total secure key), so no policy can be disadvantaged by reward shaping. We also added the switching-cost analysis and hybrid baseline requested by Reviewer 1. We believe the revised manuscript resolves the concerns and hope it now meets the bar for publication.

Reviewer 1 (Minor Revision)

We are grateful for the positive assessment and for recognizing the value of an honest negative result. We address the five points in turn.

> 1. The evaluation is limited to a standard PPO implementation with a flat MLP policy. The observed gap may partly reflect the choice of learning architecture and algorithm rather than a fundamental limitation of learning-based approaches.

Response. We agree and now state this explicitly in the new *Limitations* paragraph: our conclusions are specific to a single-algorithm, flat-MLP setting, and the gap may partly reflect that architecture rather than a fundamental limit of learning-based scheduling. We are careful throughout to frame the result as a cautionary case study for *standard* DRL, not as a refutation of learning in general, and we point to curiosity-driven exploration, hierarchical control, and learning–heuristic hybrids as the natural next steps. We also note that the new per-cost ablation (point 2) shows the shortfall persists even when the agent is optimized for each specific cost, which makes a pure tuning explanation less likely.

2. The link-switching cost is central, yet no ablation or sensitivity analysis examines how varying its magnitude affects exploration, convergence, and final performance.

Response. We added this analysis as a core new contribution (Figure 3(b) and the "Sensitivity to switching cost" paragraph). We sweep the cost over $\{0, 1, 2, 3, 4, 5\}$ minutes and, importantly, retrain a DRL agent for each value. The findings are: (i) at zero cost the DRL agent matches greedy; (ii) the gap then widens monotonically as the cost grows, and the DRL agent degrades much faster than greedy even when optimized for that exact cost; and (iii) beyond a critical cost of about five minutes, once the setup latency exceeds a typical pass duration, every policy collapses to zero because no link can be held long enough to amortize a reconnection. This both quantifies the sensitivity and explains the mechanism behind the original result.

3. The setup considers a single LEO satellite and five ground stations, which restricts insight into scalability, coordination, and congestion.

Response. We agree and now state this scope limitation explicitly in the *Limitations* paragraph and the Conclusion. The environment is in fact parameterized by the number of ground stations, and we have kept the released code configurable to support exactly the multi-station and multi-satellite studies the reviewer envisions; we flag these as the primary directions for future work rather than claim coverage we do not have.

4. Hybrid strategies combining learning and heuristics are mentioned conceptually but not evaluated empirically. Including such baselines would help contextualize the negative results.

Response. We added a cost-aware hybrid baseline: greedy selection with switching hysteresis (Eq. 9), which retains the current link unless an alternative exceeds it in elevation by a margin large enough to justify the setup latency. To avoid a strawman, we swept the hysteresis margin; it changes the result by at most 0.3%. The hybrid does not beat plain greedy, and we explain why: with one satellite and dispersed stations, the satellite is rarely in view of two ground stations at once, so most reconfigurations are forced by the held link dropping below the horizon rather than chosen to chase a marginally better one. Hysteresis can only help with discretionary switches, which are scarce here. This contextualizes the negative result and identifies the structural reason for it.

5. The paper would benefit from explicitly stating that these conclusions may not directly generalize to multi-satellite, multi-agent, or long-horizon planning scenarios.

Response. Done. The *Limitations* paragraph and the Conclusion now state plainly that the conclusions need not generalize to multi-satellite constellations, multi-agent coordination, or long-horizon planning, where learning could exhibit clearer advantages.

Reviewer 2 (Reject)

We thank the reviewer for the precise criticisms. We believe each rests on information that was missing from the original submission rather than on a flaw in the study, and we have now supplied that information.

> 1. There is insufficient information to be convinced. No information was provided on how the reward function was designed, or how much weight was placed on which elements. If the reward considers a broader range of factors than the greedy algorithm, it will inevitably score lower on certain indicators.

Response. This is the most important point and we have addressed it directly. Section 2 now specifies the complete reward as explicit equations: the free-space key-rate model (Eqs. 4–5), the switching-cost timer dynamics (Eq. 6), and the full piecewise reward (Eq. 7). Crucially, we clarify a point that was implicit before: *the reward is not a private DRL objective.* The single scalar in Eq. 7, the secure key generated net of operational penalties, is the identical objective used to score every policy, the learned agent and the greedy, hybrid, and random baselines alike. No policy optimizes or is graded on a different or more heavily weighted quantity, so the concern that the agent is penalized for considering "broader factors" does not apply: all methods are measured on the same total secure key. We have stated this explicitly in Section 2 and again where the results are reported.

> 2. The total reward in Figure 2(a) confirms convergence of the DRL algorithm. If the rewards of the three algorithms are not designed with the same formula, it is ambiguous what the graph explains.

Response. The reviewer is right to demand this, and the answer is that the rewards *are* the same formula for all policies (see the response above). To remove any remaining ambiguity, the learning curve (now Figure 3(a)) plots the DRL agent against the greedy and random baselines on this one shared objective, so the curves are directly comparable. We have also replaced the single-seed numbers with the mean and standard deviation over 20 seeds, so the comparison reflects typical rather than incidental behavior.

> 3. Why should we compare random algorithms with greedy and DRL algorithms?

Response. We have clarified this in the Experimental Setup. The random policy is a sanity-check lower bound: it confirms that the task is non-trivial and that the reward scale is meaningful. Because the random policy earns strongly negative returns once penalties are present (for example −1,427 in the Dynamic scenario), it demonstrates that the constraints genuinely bite and that the positive scores of greedy and DRL reflect real skill rather than an easy environment.

> 4. The title, abstract, and introduction mention the satellite QKD network, but why the problem is treated in a network environment is not clearly described. It is hard to understand how the experimental results affect the environment. Do you mean to suggest that heuristics like greedy are better suited than DRL in such networks?

Response. We have rewritten the motivation in the Introduction to make the network framing concrete. A single LEO satellite is a shared downlink resource that must be allocated, minute by minute, among several competing ground stations; the controller's choices determine the aggregate secure key delivered across the network, which makes this a genuine network-level scheduling problem. On the takeaway: we are careful *not* to claim that heuristics are universally better than DRL. Our claim is narrower and, we believe, more useful: for this cost-aware scheduling formulation, a standard DRL agent does not outperform a strong, domain-aware heuristic, and we identify the structural reason (geometry-forced switching) and the regime (nonzero switching cost) in which the gap appears. We present this as a benchmark and a cautionary case for practitioners, not as a general verdict.

We thank the reviewers again. We believe the revision is substantially stronger and hope it is now suitable for ACM AI Letters.